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Application of Dynamic Time Warping to Spatially Ordered Weather Data Series Clustering

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Abstract

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Acknowledgements

put acknowledgements here

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Listings

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Chapter 1

Introduction

1.0.1 Listings

You can do listings, like in Listing 1.1

Listing 1.1: Look at this cool listing. Find the rest in Appendix A.1

```
1 $ java -jar myAwesomeCode.jar
```

You can also do language highlighting for instance with Golang: And in line 6 of Listing 1.2 you can see that we can ref to lines in listings.

Listing 1.2: Hello world in Golang

```
1 package main
2
3 import "fmt"
4
5 func main() {
6     fmt.Println("hello world")
7 }
```

1.0.2 Figures

Example of a centred figure

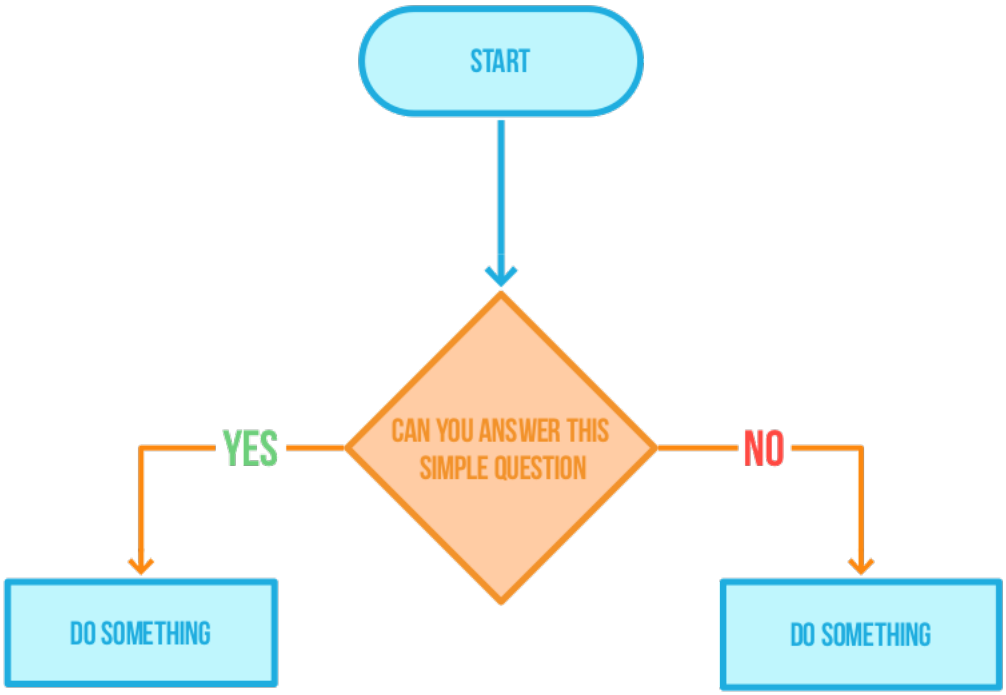


Figure 1.1: Caption for flowchart

Credit: Acme company makes everything `https://acme.com/`

1.0.3 Tables

We can also do tables. Protip: use `https://www.tablesgenerator.com/` for generating tables.

Table 1.1: Caption of table

Title1	Title2	Title3
data1	data2	data3

1.0.4 Git

Git is fun, use it!

Chapter 2

Background

2.1 Related Work

Several studies have investigated applying Dynamic Time Warping (DTW) on spatial domains and weather data separately , and other studies have investigated the decomposition of weather forecast error into spatial and temporal components . The intersection between these topics, however, is largely unexplored.

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DTW applied to spatial orderings has been used with proven results in recent litterature on automatic correlation between well-logs for geological analysis[7]. Nilo et al. use the DTW algorithm to correlate features in wells where measurements are ordered by depth, yielding promising results as an aid in research though still requiring some supervision in practical deployments [7]. Le Coz et al. use DTW together with spatial image warping to correct estimates of total rainfall during events such as convective rainstorms in sub-saharan Africa. By warping satellite-based estimates to fit measurements from local weather station networks, they note significant improvements over pure satellite-based estimates [4]. Other notable applications include using DTW to compare cloud-top temperature and other satellite-measurement time series in precipitation estimation. [5, 6]

Decomposing forecast errors into spatial and temporal components is generally first accredited to Hoffman et al. in their 1995 article 'Distortion Representation of Forecast Errors'. [2, 3] In short terms, they propose shifting weather forecast samples and scaling values to best fit the ground truth, and designating the amount of shift as displacement error, amount of scaling as amplitude error, leaving still unaccounted for error as

residual errors.[2] This has been the basis of most research and models dealing with the topic since. Jankov et al. builds on this conceptual framework to create a system that is practically applicable through the Forecast Error Decomposition method, using field alignment to morph a grid-based numerical forecast into the observed verification grid. This naturally breaks errors down into positional (displacement), structural (amplitude) and variance (residual) terms, both dealing with the double penalty problem and improving interpretability through the decomposition itself [3]. This method is also still limited to after-the-fact evaluation though. I was not able to find any literature on predictive models for uncertainty composition in forecasting, or application of the DTW algorithm on spatial orderings themselves in weather forecasting context.

2.2 Clustering

Clustering is the process of partitioning a set of data objects into groups referred to as clusters, that are similar to each other or dissimilar to objects in other clusters [1]. This is an unsupervised machine learning technique used to explore the dataset, for preprocessing or as a subroutine for further study [8]. The two main advantages of clustering is that since it is unsupervised it can easily be applied to real-world datasets that don't need to be labelled first, and that it might let us discover groupings that humans would not intuitively think of.

K-Means clustering

Glossary

Git Git is a Version Control System (VCS) for tracking changes in computer files and coordinating work on those files among multiple people.

List of Acronyms and Abbreviations

DTW Dynamic Time Warping.

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A

Generated code from Protocol buffers

Listing A.1: Source code of something

```
1 System.out.println("Hello Mars");
```

Appendix B

Statement on the use of AI tool

- Transparency Declare if and how and which AI tools were used, summarized in the Appendix, and if convenient also atappropriate points in the general text.

I am aware that I am responsible for all content of this master's thesis.